

Multibody Dynamics Primer

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The circular restricted three body problem (CR3BP) is a special case of the three body problem. In the CR3BP (much like in Keplerian 2-body dynamics), we neglect the mass of satellite S , while treating the larger celestial body m_1 and smaller celestial body m_2 as point masses. Crucially, these bodies must orbit one another in circular orbits. In other words, they both orbit about their inertially fixed barycenter c at *constant velocity and distance*.

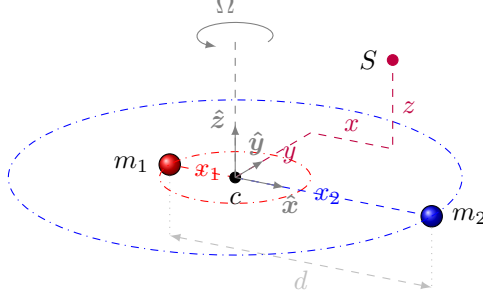


Figure 1: Geometry

The purple satellite is located by $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, and the spheres are the celestial bodies. The xyz frame is not inertially stationary. Instead, they rotate with the larger bodies at a rate Ω . The origin c is the center of mass of m_1 and m_2 (and therefore inertially fixed). The xy plane defines the plane in which m_1 and m_2 orbit, so $\hat{z}$ is defined by the direction of their angular momenta- the constant rotation at a rate Ω is a consequence of this. Note that $\mathbf{r}$ is not constrained to the xy plane, but m_1 and m_2 definitionally are.

Useful Relations

We can also find a relationship between x_1 , x_2 , d , m_1 , and m_2 . Because c is the barycenter,

$$x_1 m_1 = x_2 m_2$$

or

$$\frac{x_1}{m_2} = \frac{x_2}{m_1}$$

We can use this to find that

$$\begin{aligned} \frac{x_1}{x_1 + x_2} &= \frac{x_1/m_2}{x_1/m_2 + x_2/m_2} \\ &= \frac{x_2/m_1}{x_2/m_1 + x_2/m_2} \\ &= \frac{1/m_1}{1/m_1 + 1/m_2} \\ &= \frac{m_2}{m_2 + m_1} \end{aligned}$$

Defining $d = x_1 + x_2$ as the distance between the two celestial bodies, and $m = m_1 + m_2$ as their total mass, we get that

$$\boxed{\frac{x_1}{d} = \frac{m_2}{m}}$$

and similarly

$$\boxed{\frac{x_2}{d} = \frac{m_1}{m}}$$

Which can be rewritten as

$$\boxed{\frac{x_1}{m_2} = \frac{x_2}{m_1} = \frac{d}{m}}$$

Lastly, we will define two additional vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ which point from the first and second body respectively to the satellite.

$$\mathbf{r}_1 = (x + x_1) \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}$$

and

$$\mathbf{r}_2 = (x - x_2) \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}$$

Now we will solve for Ω .

$$\begin{aligned} \mathbf{F}_{\text{on}2} &= -\frac{Gm_1m_2}{d^2} \hat{\mathbf{x}} \\ m_2 \mathbf{a}_2^f &= -\frac{Gm_1m_2}{d^2} \hat{\mathbf{x}} \\ \frac{d^f}{dt} \left(\frac{d^f}{dt} x_2 \hat{\mathbf{x}} \right) &= -\frac{Gm_1}{d^2} \hat{\mathbf{x}} \\ \frac{d^f}{dt} \left(\frac{d^c x_2}{dt} \hat{\mathbf{x}} + \Omega \hat{\mathbf{z}} \times x_2 \hat{\mathbf{x}} \right) &= -\frac{Gm_1}{d^2} \hat{\mathbf{x}} \\ \frac{d^f}{dt} \Omega x_2 \hat{\mathbf{y}} &= -\frac{Gm_1}{d^2} \hat{\mathbf{x}} \\ \frac{d^c}{dt} \Omega x_2 \hat{\mathbf{y}} + \Omega \hat{\mathbf{z}} \times \Omega x_2 \hat{\mathbf{y}} &= -\frac{Gm_1}{d^2} \hat{\mathbf{x}} \\ -\Omega^2 x_2 \hat{\mathbf{x}} &= -\frac{Gm_1}{d^2} \hat{\mathbf{x}} \\ \Omega^2 x_2 &= \frac{Gm_1}{d^2} \\ \Omega &= \sqrt{\frac{G}{d^2} \frac{m_1}{x_2}} \\ \Omega &= \sqrt{\frac{Gm}{d^3}} \end{aligned}$$

$$\boxed{\Omega = \sqrt{\frac{Gm}{d^3}}}$$

Kinematics

The transport theorem states that the inertial (fixed f frame) derivative of a vector $\mathbf{u}$ (expressed in the rotating c frame) is

$$\frac{d^f \mathbf{u}}{dt} = \frac{d^c \mathbf{u}}{dt} + \boldsymbol{\omega}^{cf} \times \mathbf{u}$$

Where $\frac{d^f}{dt}$ denotes the derivative in the coordinates of the fixed frame f , and $\frac{d^c}{dt}$ denotes derivative in the coordinates of the rotating frame c , and $\boldsymbol{\omega}^{cf}$ denotes the angular velocity of c in f . For this case, $\boldsymbol{\omega}^{cf} = \Omega \hat{\mathbf{z}}$. We can find Ω

For the satellite's position in the CR3BP frame $\mathbf{r} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}$, we will find the inertial acceleration to generate equations of motion.

$$\begin{aligned} \dot{\mathbf{r}} &= \frac{d^c \mathbf{r}}{dt} + \boldsymbol{\omega}^{cf} \times \mathbf{r} \\ &= \frac{d^c}{dt} (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) + (\Omega \hat{\mathbf{z}} \times (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}})) \\ &= (\dot{x} \hat{\mathbf{x}} + \dot{y} \hat{\mathbf{y}} + \dot{z} \hat{\mathbf{z}}) + (\Omega x \hat{\mathbf{y}} - \Omega y \hat{\mathbf{x}}) \\ &= (\dot{x} - \Omega y) \hat{\mathbf{x}} + (\dot{y} + \Omega x) \hat{\mathbf{y}} + \dot{z} \hat{\mathbf{z}} \end{aligned}$$

$$\begin{aligned}
\ddot{\mathbf{r}} &= \dot{\mathbf{r}} \\
&= \frac{d^c}{dt} ((\dot{x} - \Omega y) \hat{\mathbf{x}} + (\dot{y} + \Omega x) \hat{\mathbf{y}} + \dot{z} \hat{\mathbf{z}}) \\
&\quad + \Omega \hat{\mathbf{z}} \times ((\dot{x} - \Omega y) \hat{\mathbf{x}} + (\dot{y} + \Omega x) \hat{\mathbf{y}} + \dot{z} \hat{\mathbf{z}}) \\
&= (\ddot{x} - \Omega \dot{y}) \hat{\mathbf{x}} + (\ddot{y} + \Omega \dot{x}) \hat{\mathbf{y}} + \ddot{z} \hat{\mathbf{z}} \\
&\quad + ((\Omega \dot{x} - \Omega^2 y) \hat{\mathbf{y}} - (\Omega \dot{y} + \Omega^2 x) \hat{\mathbf{x}}) \\
&= (\ddot{x} - 2\Omega \dot{y} - \Omega^2 x) \hat{\mathbf{x}} + (\ddot{y} + 2\Omega \dot{x} - \Omega^2 y) \hat{\mathbf{y}} + \ddot{z} \hat{\mathbf{z}} \\
\boxed{\ddot{\mathbf{r}} = (\ddot{x} - 2\Omega \dot{y} - \Omega^2 x) \hat{\mathbf{x}} + (\ddot{y} + 2\Omega \dot{x} - \Omega^2 y) \hat{\mathbf{y}} + \ddot{z} \hat{\mathbf{z}}}
\end{aligned}$$

Or, rewritten,

$$\begin{aligned}
\ddot{\mathbf{r}} &= (\ddot{x} - 2\Omega \dot{y} - \Omega^2 x) \hat{\mathbf{x}} + (\ddot{y} + 2\Omega \dot{x} - \Omega^2 y) \hat{\mathbf{y}} + \ddot{z} \hat{\mathbf{z}} \\
\ddot{\mathbf{r}} &= (\ddot{x} \hat{\mathbf{x}} + \ddot{y} \hat{\mathbf{y}} + \ddot{z} \hat{\mathbf{z}}) - \Omega^2 (x \hat{\mathbf{x}} + y \hat{\mathbf{y}}) - 2\Omega \dot{y} \hat{\mathbf{x}} + 2\Omega \dot{x} \hat{\mathbf{y}} \\
\ddot{\mathbf{r}} &= \mathbf{a}^c - \Omega^2 \mathbf{r}_{xy} + 2\Omega \times \mathbf{v}^c
\end{aligned}$$

$\ddot{\mathbf{r}} = \underbrace{\mathbf{a}^c}_{\text{Relative Acceleration}} + \overbrace{-\Omega^2 \mathbf{r}_{xy}}^{\text{Centripetal Acceleration}} + \underbrace{2\Omega \times \mathbf{v}^c}_{\text{Coriolis Acceleration}}$
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Equations of Motion

We can now generate the equations of motion

$$\begin{aligned}
\sum_i \mathbf{F}_i &= m \ddot{\mathbf{r}} \\
\mathbf{F}_1 + \mathbf{F}_2 &= m \ddot{\mathbf{r}} \\
-\frac{Gm_1 m}{r_1^3} \mathbf{r}_1 - \frac{Gm_2 m}{r_2^3} \mathbf{r}_2 &= m \ddot{\mathbf{r}} \\
-\frac{Gm_1}{r_1^3} \mathbf{r}_1 - \frac{Gm_2}{r_2^3} \mathbf{r}_2 &= (\ddot{x} - 2\Omega \dot{y} - \Omega^2 x) \hat{\mathbf{x}} + (\ddot{y} + 2\Omega \dot{x} - \Omega^2 y) \hat{\mathbf{y}} + \ddot{z} \hat{\mathbf{z}}
\end{aligned}$$

We now write this as three equations, one each in x , y , and z

$$\begin{aligned}
-\frac{Gm_1}{r_1^3}(x + x_1) - \frac{Gm_2}{r_2^3}(x - x_2) &= \ddot{x} - 2\Omega \dot{y} - \Omega^2 x \\
-\frac{Gm_1}{r_1^3}y - \frac{Gm_2}{r_2^3}y &= \ddot{y} + 2\Omega \dot{x} - \Omega^2 y \\
-\frac{Gm_1}{r_1^3}z - \frac{Gm_2}{r_2^3}z &= \ddot{z}
\end{aligned}$$

Isolating the second derivatives,

$$\begin{aligned}
\ddot{x} &= -\frac{Gm_1}{r_1^3}(x + x_1) - \frac{Gm_2}{r_2^3}(x - x_2) + 2\Omega \dot{y} + \Omega^2 x \\
\ddot{y} &= -\frac{Gm_1}{r_1^3}y - \frac{Gm_2}{r_2^3}y - 2\Omega \dot{x} + \Omega^2 y \\
\ddot{z} &= -\frac{Gm_1}{r_1^3}z - \frac{Gm_2}{r_2^3}z
\end{aligned}$$

We can now substitute $\Omega = \sqrt{\frac{Gm}{d^3}} = \sqrt{\frac{Gm_1 + Gm_2}{d^3}}$

$$\begin{aligned}
\ddot{x} &= -\frac{Gm_1}{r_1^3}(x+x_1) - \frac{Gm_2}{r_2^3}(x-x_2) + 2\sqrt{\frac{Gm}{d^3}}\dot{y} + \frac{Gm}{d^3}x \\
\ddot{y} &= -\frac{Gm_1}{r_1^3}y - \frac{Gm_2}{r_2^3}y - 2\sqrt{\frac{Gm}{d^3}}\dot{x} + \frac{Gm}{d^3}y \\
\ddot{z} &= -\frac{Gm_1}{r_1^3}z - \frac{Gm_2}{r_2^3}z
\end{aligned}$$

Nondimensional Equations of Motion

We will now begin to non-dimensionalize the EOMs.

First, we will change the time unit. We will pick $t^* = t\sqrt{Gm/d^3}$ (recall that $\sqrt{Gm/d^3} = \Omega$, which has dimensions of time^{-1}). Everywhere that a derivative is present, it is implied to be with respect to the time unit of 1 second. We must therefore switch from implied $\frac{d}{dt}$ to implied $\frac{d}{dt^*}$.

$$\begin{aligned}
\frac{d(\quad)}{dt} &= \frac{d(\quad)}{dt^*} \frac{dt^*}{dt} \\
&= \frac{d(\quad)}{dt^*} \sqrt{\frac{Gm}{d^3}}
\end{aligned}$$

From this, it can be seen that to make a derivative with time implied to be with the nondimensional time, $\sqrt{d^3/Gm}$ must be multiplied for each derivative taken. The EOMs can now be rewritten this way, with dots implied to be relative to the nondimensional time unit

$$\begin{aligned}
\left(\frac{d^2x}{dt^{*2}} \frac{Gm}{d^3}\right) &= -\frac{Gm_1}{r_1^3}(x+x_1) - \frac{Gm_2}{r_2^3}(x-x_2) + 2\sqrt{\frac{Gm}{d^3}} \left(\frac{dy}{dt^*} \sqrt{\frac{Gm}{d^3}}\right) + \frac{Gm}{d^3}x \\
\left(\frac{d^2y}{dt^{*2}} \frac{Gm}{d^3}\right) &= -\frac{Gm_1}{r_1^3}y - \frac{Gm_2}{r_2^3}y - 2\sqrt{\frac{Gm}{d^3}} \left(\frac{dx}{dt^*} \sqrt{\frac{Gm}{d^3}}\right) + \frac{Gm}{d^3}y \\
\left(\ddot{z} \frac{Gm}{d^3}\right) &= -\frac{Gm_1}{r_1^3}z - \frac{Gm_2}{r_2^3}z
\end{aligned}$$

Some algebraic simplifications can now be made

$$\begin{aligned}
\frac{d^2x}{dt^{*2}} &= -\frac{m_1d^3}{mr_1^3}(x+x_1) - \frac{m_2d^3}{mr_2^3}(x-x_2) + 2\frac{dy}{dt^*} + x \\
\frac{d^2y}{dt^{*2}} &= -\frac{m_1d^3}{mr_1^3}y - \frac{m_2d^3}{mr_2^3}y - 2\frac{dx}{dt^*} + y \\
\frac{d^2z}{dt^{*2}} &= -\frac{m_1d^3}{mr_1^3}z - \frac{m_2d^3}{mr_2^3}z
\end{aligned}$$

Next, we can define nondimensional distances and masses. The relationship between a nondimensional distance L^* and its dimensional counterpart is $L^* = L/d$. Furthermore, we define nondimensional masses a similar relationship $M^* = M/m$. With this defined, we can make some substitutions in the EOMs.

$$\begin{aligned}
\frac{d^2x}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}(x+x_1) - \frac{m_2^*}{r_2^{*3}}(x-x_2) + 2\frac{dy}{dt^*} + x \\
\frac{d^2y}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}y - \frac{m_2^*}{r_2^{*3}}y - 2\frac{dx}{dt^*} + y \\
\frac{d^2z}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}z - \frac{m_2^*}{r_2^{*3}}z
\end{aligned}$$

By dividing both sides by d , the remaining distances can be made nondimensional.

$$\begin{aligned}\frac{d^2 x^*}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}(x^* + x_1^*) - \frac{m_2^*}{r_2^{*3}}(x^* - x_2^*) + 2\frac{dy^*}{dt^*} + x^* \\ \frac{d^2 y^*}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}y^* - \frac{m_2^*}{r_2^{*3}}y^* - 2\frac{dx^*}{dt^*} + y^* \\ \frac{d^2 z^*}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}z^* - \frac{m_2^*}{r_2^{*3}}z^*\end{aligned}$$

The final set of substitutions will now be made: $m_2 = m - m_1 \rightarrow m_2^* = 1 - m_1^*$. Recall from the useful relations that $\frac{x_1}{d} = \frac{m_2}{m}$, Therefore $x_1^* = 1 - m_1^*$ (and $x_2^* = m_1^*$). While r_1^* and r_2^* can be written in terms of x_1^* , x^* , y^* , and z^* , I will not do this as it makes the equations much less compact.

$$\begin{aligned}\frac{d^2 x^*}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}(x^* + 1 - m_1^*) - \frac{1 - m_1^*}{r_2^{*3}}(x^* - m_1^*) + 2\frac{dy^*}{dt^*} + x^* \\ \frac{d^2 y^*}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}y^* - \frac{1 - m_1^*}{r_2^{*3}}y^* - 2\frac{dx^*}{dt^*} + y^* \\ \frac{d^2 z^*}{dt^{*2}} &= -\frac{m_1^*}{r_1^{*3}}z^* - \frac{1 - m_1^*}{r_2^{*3}}z^*\end{aligned}$$

I will write this with the nondimensionalization implicit instead of explicit. In this space, all parameters represent their nondimensional counterpart.

$$\begin{aligned}\ddot{x} &= -\frac{m_1}{r_1^3}(x + 1 - m_1) - \frac{1 - m_1}{r_2^3}(x - m_1) + 2\dot{y} + x \\ \ddot{y} &= -\frac{m_1}{r_1^3}y - \frac{1 - m_1}{r_2^3}y - 2\dot{x} + y \\ \ddot{z} &= -\frac{m_1}{r_1^3}z - \frac{1 - m_1}{r_2^3}z\end{aligned}$$

If instead m_2 is used as the characterizing parameter (rather than m_1), the EOMs change slightly. Conventionally, this parameter is written μ (not to be confused with the gravitational parameter $\mu = G(m_1 + m_2)$)

$$\begin{aligned}\ddot{x} &= -\frac{1 - \mu}{r_1^3}(x + \mu) - \frac{\mu}{r_2^3}(x - 1 + \mu) + 2\dot{y} + x \\ \ddot{y} &= -\frac{1 - \mu}{r_1^3}y - \frac{\mu}{r_2^3}y - 2\dot{x} + y \\ \ddot{z} &= -\frac{1 - \mu}{r_1^3}z - \frac{\mu}{r_2^3}z\end{aligned}$$

Noting that

$$\begin{aligned}r_1 &= \sqrt{(x + \mu)^2 + y^2 + z^2} \\ r_2 &= \sqrt{(x - 1 + \mu)^2 + y^2 + z^2}\end{aligned}$$

The solutions to this differential equation solve the non-dimensional CR3BP, in which the bodies' masses add to 1 and the distance between them is 1. To translate it into real solutions, the time scale, distance scale, and mass scale must be applied. Therefore the only thing in this equation that depends upon the specific two-body system of interest is m_1 , or the fraction of the primary mass to the total masses of both planetary.

Energy Analysis

To find the energy of the system, we must find a potential function U for the fictitious centrifugal acceleration $-\Omega^2 (x\hat{x} + y\hat{y}) = -\frac{Gm}{d^3} (x\hat{x} + y\hat{y})$. The potential function that satisfies this is

$$U = -\frac{1}{2} \frac{Gm}{d^3} (x^2 + y^2)$$

We can analyse the energy of a CR3BP system to determine possible positions. The total energy of the system is.

$$\begin{aligned} E &= \frac{1}{2} v_{\text{in CR3BP frame}}^2 + \overbrace{U_{\text{grav}} + U_{\text{centrifugal}}}^{\text{Pseudopotential}} \\ &= \frac{1}{2} (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - \left(\frac{Gm_1}{r_1} + \frac{Gm_2}{r_2} \right) - \frac{1}{2} \frac{Gm}{d^3} (x^2 + y^2) \end{aligned}$$

This can be nondimensionalized to

$$\frac{1}{2} (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - \underbrace{\left(\frac{1-\mu}{r_1} + \frac{\mu}{r_2} \right) - \frac{1}{2} (x^2 + y^2)}_{\text{Pseudopotential}}$$

This is a conserved quantity. Different sources treat it differently, with some doubling or negating it. In the interest of aligning with JPL's Three Body Periodic Orbits Database [1], I will double and negate this energy; this value is called the Jacobi constant (or Jacobi integral)

$$J = 2 \left(\frac{1-\mu}{r_1} + \frac{\mu}{r_2} \right) + (x^2 + y^2) - (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

The zero-velocity contours are given by

$$J = 2 \left(\frac{1-\mu}{r_1} + \frac{\mu}{r_2} \right) + (x^2 + y^2) = -2U_{\text{pseudo}}(x, y, z, \mu)$$

Keeping in mind that r_1 and r_2 are functions of x, y, z, μ . These curves can give some intuition for regions that a spacecraft cannot enter given a quantity of energy.

In the following figures, "colder" colors (towards blue) correspond to lower Jacobi constants, while "warmer" colors (yellows, oranges, and towards red) correspond to higher values. A spacecraft with a given amount of energy is bounded outside of any region encompassed by a warmer color. For example, a spacecraft with some amount of energy corresponding to green cannot enter any area with colors warmer than green. Put another way, colder colors are lower on the pseudopotential surface than are warmer colors.

Figure 2: Level Curves of the pseudopotential for $\mu = 0.01$

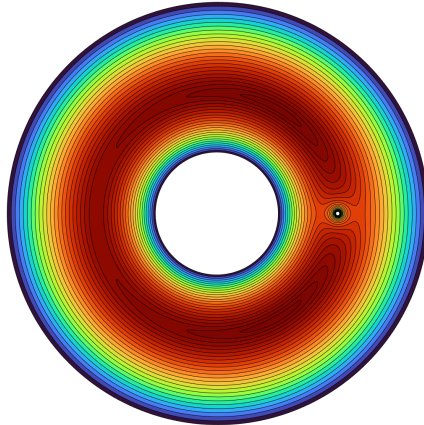


Figure 3: Level Curves of the pseudopotential for $\mu = 0.1$

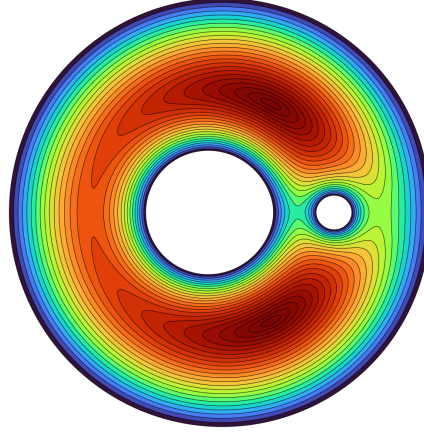


Figure 4: Level Curves of the pseudopotential for $\mu = 0.25$

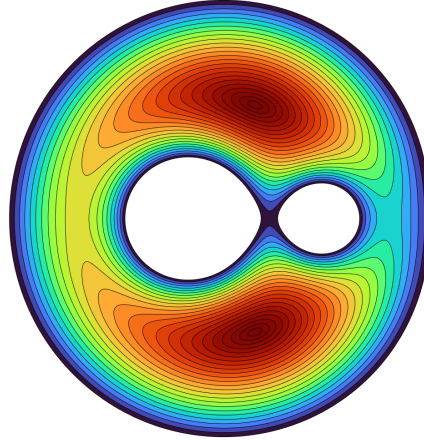
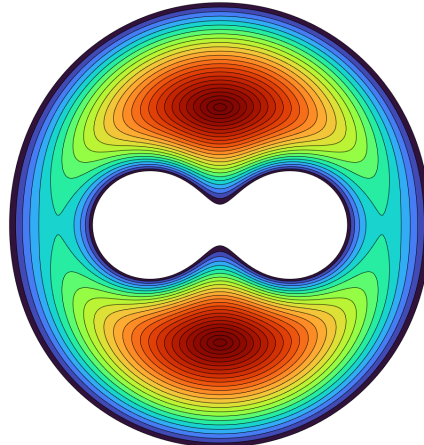


Figure 5: Level Curves of the pseudopotential for $\mu = 0.5$



For reference, the mass ratio value μ for the Earth-Moon system is $\mu \approx 0.012$, which is roughly that of the first diagram. Clearly, entering regions near the orbit of the second body is challenging, but exceeding those regions is quite feasible if done in the proximity of the secondary body.

Zero-Velocity Solutions - Lagrange Points

There are some solutions for which the velocity is zero for all times, called Lagrange Points or libration points. To find these, we simply set the derivatives in the EOMs equal to zero.

$$\begin{aligned}\dot{x} &= -\frac{1-\mu}{r_1^3}(x+\mu) - \frac{\mu}{r_2^3}(x-1+\mu) + 2\mathcal{J} + x \\ \dot{y} &= -\frac{1-\mu}{r_1^3}y - \frac{\mu}{r_2^3}y - 2\mathcal{K} + y \\ \dot{z} &= -\frac{1-\mu}{r_1^3}z - \frac{\mu}{r_2^3}z\end{aligned}$$

Looking at the z EOM, the right-hand-side must have the opposite sign to z . Therefore, for the velocity to remain zero, $z = 0 \forall t$. We will examine the case of $y = 0$ and $y \neq 0$ separately.

First, we look at $y = 0$, which eliminates the second equality above.

$$\begin{aligned}\dot{x} &= -\frac{1-\mu}{r_1^3}(x+\mu) - \frac{\mu}{r_2^3}(x-1+\mu) + 2\mathcal{J} + x \\ 0 &= -\frac{1-\mu}{\sqrt{(x+\mu)^2}^3}(x+\mu) - \frac{\mu}{\sqrt{(x-1+\mu)^2}^3}(x-1+\mu) + x \\ 0 &= -\frac{1-\mu}{|x+\mu|^3}(x+\mu) - \frac{\mu}{|x-1+\mu|^3}(x-1+\mu) + x\end{aligned}$$

This has three solutions, typically called L_1 (the solution between the two bodies), L_2 (the solution on the far side of the smaller body), and L_3 (the solution on the far side of the larger body).

Now, we examine $y \neq 0$

$$\begin{aligned}\dot{y} &= -\frac{1-\mu}{r_1^3}y - \frac{\mu}{r_2^3}y - 2\mathcal{K} + y \\ 0 &= -\frac{1-\mu}{r_1^3}y - \frac{\mu}{r_2^3}y + y \\ 0 &= -\frac{1-\mu}{r_1^3} - \frac{\mu}{r_2^3} + 1 \\ 1 &= \frac{1-\mu}{r_1^3} + \frac{\mu}{r_2^3} \\ \frac{1-\mu}{r_1^3} &= 1 - \frac{\mu}{r_2^3}\end{aligned}$$

Bringing in the equation for x ,

$$\begin{aligned}0 &= -\frac{1-\mu}{r_1^3}(x+\mu) - \frac{\mu}{r_2^3}(x-1+\mu) + x \\ 0 &= -\left(1 - \frac{\mu}{r_2^3}\right)(x+\mu) - \frac{\mu}{r_2^3}(x-1+\mu) + x \\ 0 &= -(x+\mu) + \frac{\mu}{r_2^3}(x+\mu) - \frac{\mu}{r_2^3}(x-1+\mu) + \frac{\mu}{r_2^3} + x \\ 0 &= -x - \mu + \frac{\mu}{r_2^3} + x \\ 0 &= -\mu + \frac{\mu}{r_2^3} \\ \frac{1}{r_2^3} &= 1 \\ r_2 &= 1\end{aligned}$$

Plugging this back into $\frac{1-\mu}{r_1^3} = 1 - \frac{\mu}{r_2^3}$,

$$\begin{aligned}\frac{1-\mu}{r_1^3} &= 1 - \frac{\mu}{r_2^3} \\ \frac{1-\mu}{r_1^3} &= 1 - \mu \\ \frac{1}{r_1^3} &= 1 \\ r_1 &= 1\end{aligned}$$

The conditions $r_1 = r_2 = 1$ locate the final two Lagrange points, forming equilateral triangles with the line connecting the two bodies.

$$\begin{aligned}r_1 &= r_2 \\ \sqrt{(x+\mu)^2 + y^2} &= \sqrt{(x-1+\mu)^2 + y^2} \\ (x+\mu)^2 + y^2 &= (x-1+\mu)^2 + y^2 \\ (x+\mu)^2 &= (x-1+\mu)^2 \\ x^2 + \mu^2 + 2x\mu &= x^2 - 2x + 2x\mu + 1 - 2\mu + \mu^2 \\ 0 &= -2x + 1 - 2\mu \\ 2x &= 1 - 2\mu \\ x &= \frac{1}{2} - \mu\end{aligned}$$

Plugging this once again into $r_1 = \sqrt{(x+\mu)^2 + y^2} = 1$,

$$\begin{aligned}\sqrt{(x+\mu)^2 + y^2} &= 1 \\ \left(\frac{1}{2} - \mu + \mu\right)^2 + y^2 &= 1 \\ \frac{1}{4} + y^2 &= 1 \\ y^2 &= \frac{3}{4} \\ y &= \pm \frac{\sqrt{3}}{2}\end{aligned}$$

The points $y = \pm\sqrt{3}/2$, $x = \frac{1}{2} - \mu$ locate L_4 and L_5 .
The Lagrange points are therefore

Point	x	y
L_1	Numeric Solution (see below)	0
L_2	Numeric Solution (see below)	0
L_3	Numeric Solution (see below)	0
L_4	$\frac{1}{2} - \mu$	$\frac{\sqrt{3}}{2}$
L_5	$\frac{1}{2} - \mu$	$\frac{\sqrt{3}}{2}$

$$L_{1,2,3} : -\frac{1-\mu}{|x_{L1,2,3} + \mu|^3}(x+\mu) - \frac{\mu}{|x-1+\mu|^3}(x-1+\mu) + x = 0$$

The Lagrange points are shown below for the Earth-Moon system ($\mu = 0.01215$) as well as for the Pluto-Charon system ($\mu = 0.10828$)0829

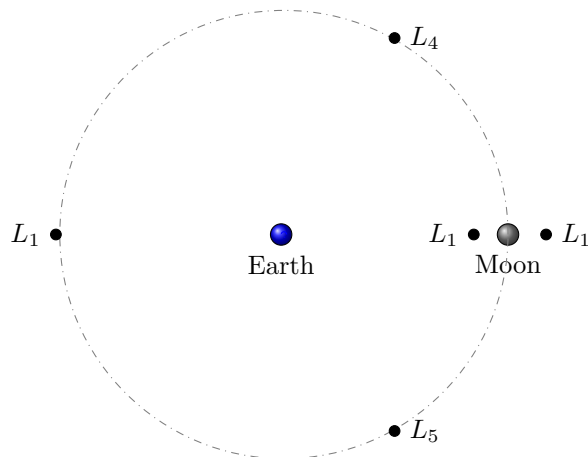


Figure 6: Earth-Moon Lagrange Points. Earth-Moon barycenter is within Earth

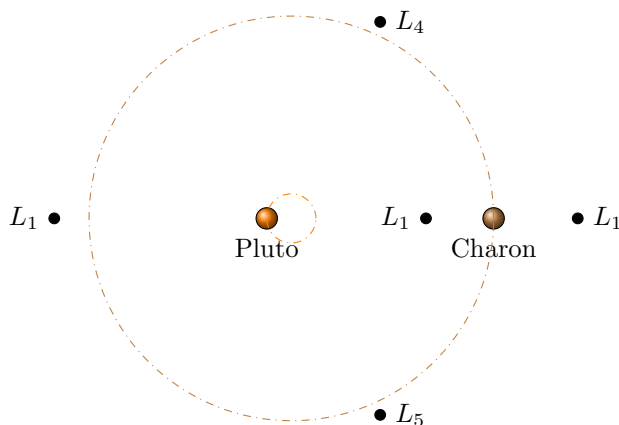


Figure 7: Pluto-Charon Lagrange Points

CR3BP Canonical Dynamics

Now that you are familiar with the governing equations of CR3BP dynamics, canonical behavior can be explored. First, a qualitative note:

Motion in the rotating frame acts as "feels" natural under gravitational influence, except for two effects derived earlier in this document.

1. Centripetal acceleration causes objects to accelerate outward proportionally to their distance from the barycenter.
2. The Coriolis acceleration causes objects to curve counterclockwise (when viewed from above) proportionally to their velocity.

With what little intuition that can be established having been addressed, canonical families of orbits in the CR3BP can be examined.

In Keplerian mechanics, the space of all possible orbits is five-dimensional¹ with few degenerate cases (such as circular or equatorial orbits) that remove dimensions. Furthermore,

¹With an orbit being uniquely determined by semimajor axis, inclination, eccentricity, right ascension of ascending node, and argument of periapsis true anomaly locates the body along its orbit

all initial conditions in Keplerian dynamics lead to periodic solutions[2]. This is not the case for CR3BP dynamics. The constraint of periodicity enforces many requirements on initial conditions, leading to a single-dimensional continuum of periodic orbits where any exist at all [Note to author: this should be cited at some point]. Conventionally, these orbits are broken into a few different "families".

1 Families of Trajectories

There are a plethora of families of trajectories within CR3BP; infinitely many, in fact[3]. There are many methods of finding them. For the sake of orbital analysis, JPL's three-body-orbit browser likely suffices. However, I will describe natural parameter continuation and differential corrections.

1.1 Linear Dynamics

To find a solution near a libration point, linearized dynamics can be used and linear solutions can be corrected.

The linearized dynamics is can be written[3] as

$$A = \begin{bmatrix} \mathcal{L} & \mathbb{I} \\ U_{xx} & \Omega \end{bmatrix}$$

Where

$$\Omega = \begin{bmatrix} 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

And U_{xx} is the Hessian of energy.

The linear solution is then

$$\mathbf{x}(t) = \sum_{j=1}^6 C_j e^{\lambda_j t}$$

From numerical methods, A has only two purely imaginary eigenvalues (required for a periodic solution), which will be denoted $\pm s$. The linear solution therefore has $x(t) = C_1 \cos(st) + C_2 \sin(st)$ and $y(t) = C_3 \cos(st) + C_4 \sin(st)$. It can be shown [AAE632 notes] that $C_3 = iC_1 \frac{s^2 + U_{xx}}{2s}$ and $C_4 = -iC_2 \frac{s^2 + U_{xx}}{2s}$. This this means that for $y(0) = 0$, we have $C_1 = -C_2$, and from $x(t)$ we find $C_1 = \Delta$ where Δ is the stepoff from the Lagrange point, This gives $\dot{y}_0 = -\frac{s^2 + U_{xx}}{2} \Delta$

The linear solution is of course wrong in the nonlinear system, so a differential corrector scheme is utilized to seek symmetry across the xz plane.

1.2 Differential Corrections

First, we propagate to the first xz -plane crossing. If the change in propagation time δt_f is appended to the state, the STM (State Transition Matrix; maps initial deviations to final deviations) becomes

$$\Phi' = \begin{bmatrix} \Phi_{rr} & \Phi_{vv} & \mathbf{v} \\ \Phi_{vr} & \Phi_{vv} & \mathbf{a} \end{bmatrix}$$

Where the prime is used because it's not a "real" state transition matrix. Propagating to the plane then gives us

$$\begin{bmatrix} \delta x \\ \delta y \\ \delta z \\ \delta \dot{x} \\ \delta \dot{y} \\ \delta \dot{z} \end{bmatrix}_f = \begin{bmatrix} \Phi_{rr} & \Phi_{vv} & \mathbf{v} \\ \Phi_{vr} & \Phi_{vv} & \mathbf{a} \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \\ \delta z \\ \delta \dot{x} \\ \delta \dot{y} \\ \delta \dot{z} \\ \delta t \end{bmatrix}_0$$

Our design variable is only $\delta \dot{y}_0$ and δt , so all else can be dropped from the right hand side. On the left hand side, we are propagating to the plane, setting δy_f to zero, and targeting $\delta \dot{x}_f = 0$ for a perpendicular crossing. This gives

$$\begin{bmatrix} \delta y_f \\ \delta \dot{x}_f \end{bmatrix} = \begin{bmatrix} \frac{\partial y_f}{\partial \dot{y}_0} & \frac{\partial y_f}{\partial t} \\ \frac{\partial \dot{x}_f}{\partial \dot{y}_0} & \frac{\partial \dot{x}_f}{\partial t} \end{bmatrix} \begin{bmatrix} \delta \dot{y}_0 \\ \delta t \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ \delta \dot{x}_f \end{bmatrix} = \begin{bmatrix} \phi_{25} & \dot{y}_f \\ \phi_{45} & \dot{x}_f \end{bmatrix} \begin{bmatrix} \delta \dot{y}_0 \\ \delta t \end{bmatrix}$$

Looking at just the first row, we have

$$0 = \phi_{25} \delta \dot{y}_0 + \dot{y}_f \delta t$$

$$\dot{y}_f \delta t = -\phi_{25} \delta \dot{y}_0$$

$$\delta t = -\phi_{25} \frac{\delta \dot{y}_0}{\dot{y}_f}$$

Now, expanding the second row,

$$\delta \dot{x}_f = \phi_{45} \delta \dot{y}_0 + \ddot{x}_f \delta t$$

$$= \phi_{45} \delta \dot{y}_0 + \ddot{x}_f \left(-\phi_{25} \frac{\delta \dot{y}_0}{\dot{y}_f} \right)$$

$$= \left(\phi_{45} \delta \dot{y}_0 - \phi_{25} \frac{\ddot{x}_f}{\dot{y}_f} \right) \delta \dot{y}_0$$

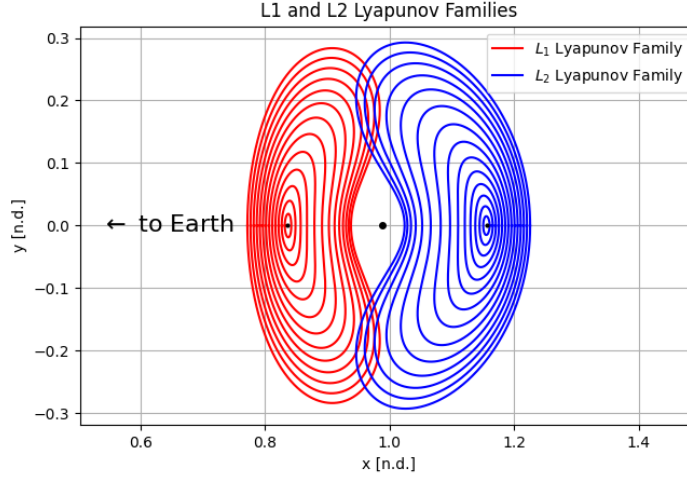
We want $\delta \dot{x}_f = -\dot{x}_{f,actual}$ (targetting $\dot{x}_f = 0$), so the differential correction becomes

$$\delta \dot{y}_0 = \frac{-\dot{x}_{f,actual}}{\left(\phi_{45} \delta \dot{y}_0 - \phi_{25} \frac{\ddot{x}_f}{\dot{y}_f} \right)}$$

This gives rise to a new trial initial condition, which is then propagated again to the xz -plane crossing, and another differential correction is applied. This is repeated until the correction is within some tolerance. While this process works near the Lagrange point, a sufficiently large step away from the Lagrange point would yeild a fully invalid linear solution.

This family of solutions is called the Lyapunov family:

Figure 8: L_1 and L_2 Lyapunov Families



1.3 Bifurcation To Halo

If the STM is propagated for a full orbit, it will have six eigenvalues (since $\Phi \in \mathbb{R}^{6 \times 6}$). Two eigenvalues will be unity (the unity eigenspace goes both along the trajectory and in the direction of the next trajectory). The other four eigenvalues occur in conjugate pairs. When a second trajectory can be continued from a reference trajectory, the new trajectory will undergo a qualitative change from the first. This is a bifurcation, and it can be found when another eigenvalue pair approaches 1.

Lyapunovs bifurcate to the Halo family, which exhibit out-of-plane behavior. The Halo family includes trajectories that, in the rotating frame, appear nearly planar. These are the Near Rectilinear Halo Orbits (NRHOs).

Once the bifurcation point is found, an out-of-plane targetter can be used to find members of this family. Various targetting schemes exist to find new orbits in a family.

Figure 9: L_1 Halo Family (with some NRHOs)
L1 Halo Family

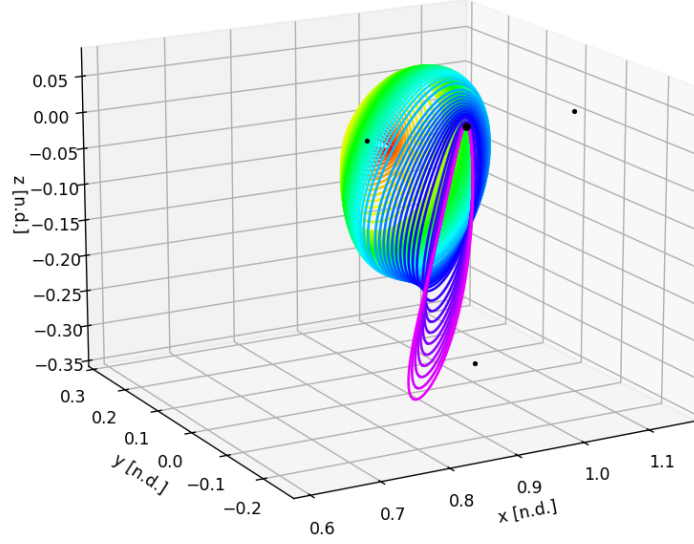
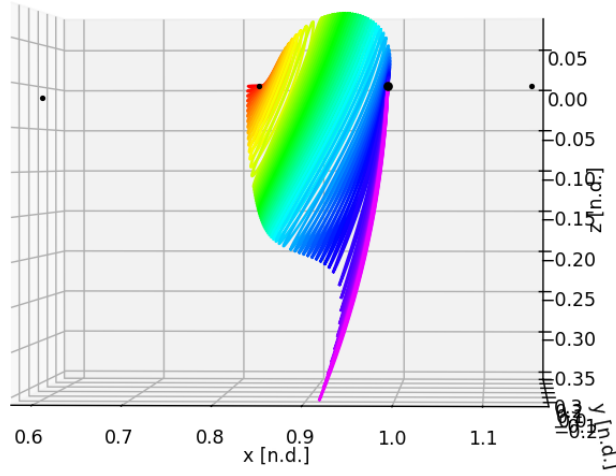


Figure 10: L_1 Halo Family (with some NRHOs) - View From $-y$



2 Manifolds and Transfers

Unstable trajectories feature hyperbolic manifolds; these are structures that unstably perturbed points along an orbit will follow. Due to the chaos inherent to the CR3BP dynamical system, these manifolds often reach quite far in the rotating frame.

Figure 11: L_1 Halo panifolds propagated for 10 TU

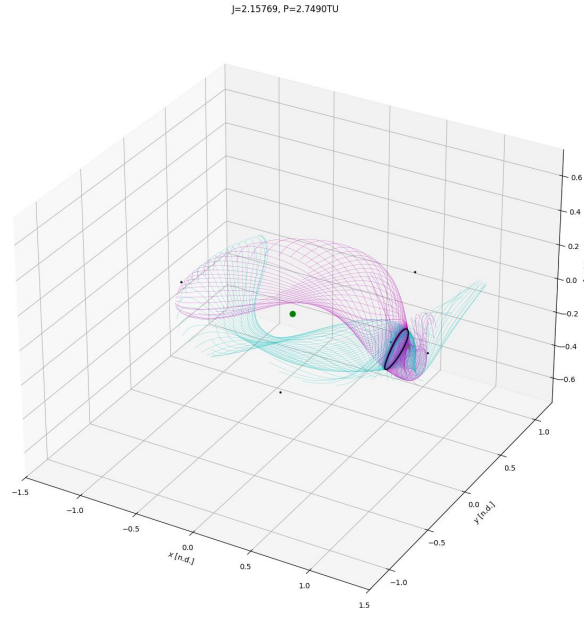


Figure 12: L_1 Lyapunov manifolds propagated to xz -crossing

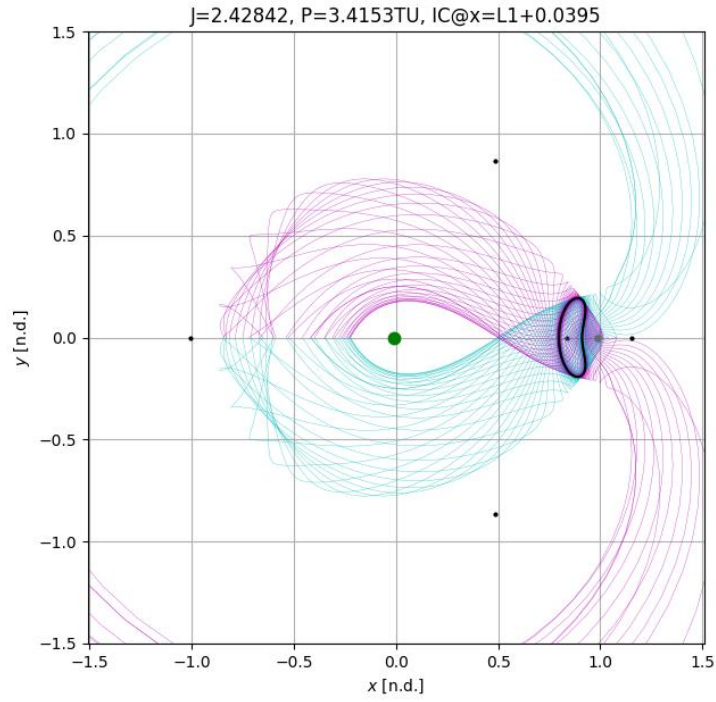
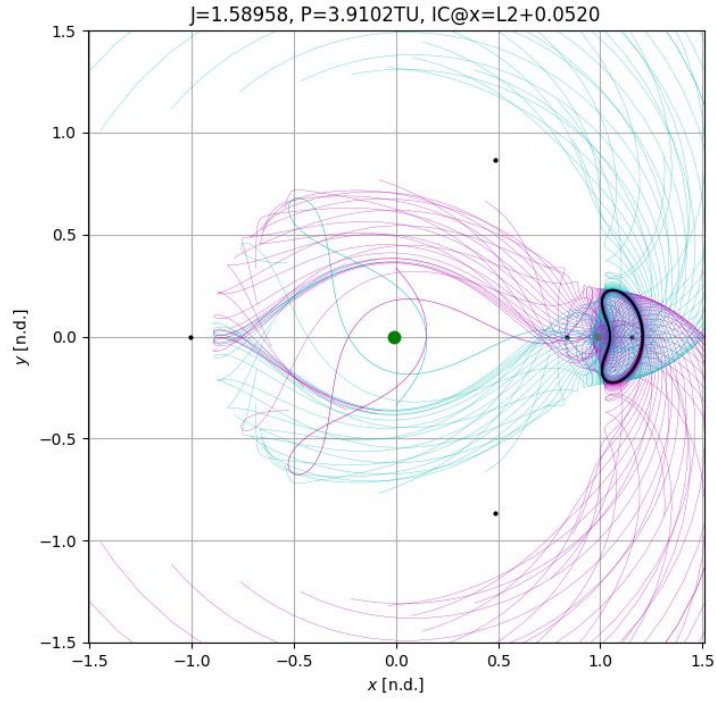


Figure 13: L_2 Lyapunov manifolds propagated for 2 periods



In the above figures, the titles show Jacobi constants, periods, and initial conditions. The stable and unstable manifolds are both shown. The stable manifold asymptotically approaches the trajectory, while the unstable asymptotically departs it. Cheap transfers can be found by finding points of intersection between the unstable manifolds of one orbit and the stable manifolds of another.

Note that I consulted many references during the making of this. Where applicable, specific references are cited.

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